

[illegible]

1. Beta-blocker first-line

WHAT YOU SEE

'BETA BLOCKER' shield/banner at top; beta-blocker soldier

WHAT IT ENCODES

Beta-blockers (metoprolol succinate 25–200 mg/day, atenolol, bisoprolol, carvedilol) are FIRST-LINE antianginal therapy: by blocking beta-1 receptors they lower heart rate and contractility, reducing myocardial oxygen demand ($\downarrow$ MVO₂), and the slower HR lengthens diastole to improve coronary perfusion. Titrate to a resting HR of ~55–60 bpm. They are the only antianginal class with proven mortality benefit post-MI and in HFrEF, so they are preferred when a comorbidity (prior MI, low EF) coexists.

BOARD TRAP

WRONG: a calcium channel blocker or nitrate is the first-line antianginal of choice in stable angina. Discriminator: beta-blockers are first-line (and uniquely lower mortality post-MI/in HFrEF); CCBs and nitrates are add-ons or alternatives when beta-blockers are contraindicated or insufficient.

2. BB abrupt stop → rebound

WHAT YOU SEE

Beta-blocker soldier 'ABRUPT STOP', calendar 'REBOUND $\uparrow$ HR $\uparrow$ MVO₂', 2-week taper

WHAT IT ENCODES

Never stop beta-blockers abruptly in IHD: chronic blockade upregulates beta-receptors, so sudden withdrawal causes rebound tachycardia, hypertension, and $\uparrow$ MVO₂ that can precipitate angina, MI, or arrhythmia. Taper gradually over ~1–2 weeks. This rebound phenomenon is also why beta-blockers should generally be continued perioperatively rather than discontinued.

BOARD TRAP

WRONG: beta-blockers can be stopped abruptly without consequence (e.g., before surgery). Discriminator: abrupt withdrawal triggers rebound tachycardia/ischemia and can precipitate MI — taper over 1–2 weeks; this withdrawal hazard is a favorite exam point.

3. Long-acting nitrate

WHAT YOU SEE

'LONG-ACTING NITRATE POUCH', nitrate knight

WHAT IT ENCODES

Long-acting nitrates (isosorbide mononitrate, isosorbide dinitrate, transdermal nitroglycerin patch) are add-on antianginals. Nitrates donate NO → venodilation that reduces preload and ventricular wall stress (lowering MVO₂), with some epicardial coronary dilation. They relieve symptoms but have NO mortality benefit. Add when a beta-blocker alone is insufficient or as initial therapy if beta-blockers are contraindicated.

BOARD TRAP

WRONG: long-acting nitrates reduce mortality in stable IHD. Discriminator: nitrates are purely symptomatic (no survival benefit) — only beta-blockers, and revascularization in select anatomy, affect mortality; nitrates are for angina relief.

4. Nitrate-free interval

WHAT YOU SEE

Clock '10–12 HOUR NITRATE-FREE INTERVAL', 'TOLERANCE'

WHAT IT ENCODES

Continuous nitrate exposure causes tolerance (tachyphylaxis) within ~24h via depleted sulfhydryl groups / oxidative stress, abolishing efficacy. Prevent it by building in a daily nitrate-free interval of 10–14 hours (e.g., remove the patch overnight, dose isosorbide mononitrate once daily / dinitrate in an asymmetric schedule). The nitrate-free window is usually scheduled at night when angina risk is lowest.

BOARD TRAP

WRONG: dosing long-acting nitrates around the clock for steady coverage is optimal. Discriminator: continuous dosing causes tolerance and loss of effect — you must include a daily 10–14h nitrate-free interval; round-the-clock nitrates defeat the therapy.

5. Sublingual NTG (acute)

WHAT YOU SEE

'SUBLINGUAL TABLET / transdermal NTG patch', nitrate knight

WHAT IT ENCODES

Sublingual nitroglycerin (0.3–0.4 mg tablet or spray) is for ACUTE angina relief or prophylaxis just before exertion; onset is 1–3 min. Instruct: one dose, repeat every 5 min up to 3 doses; if pain persists after the first dose, call emergency services. Absolute contraindication: recent PDE5 inhibitor use (sildenafil/tadalafil/vardenafil) — the combination causes life-threatening hypotension.

BOARD TRAP

WRONG: sublingual nitroglycerin is safe to give a patient who recently took sildenafil for chest pain. Discriminator: nitrate + PDE5 inhibitor → profound, potentially fatal hypotension — always ask about sildenafil/tadalafil before any nitrate (sublingual or IV).

6. DHP CCB (amlodipine)

WHAT YOU SEE

'LONG-ACTING DHP = SAFE', amlodipine 'DHP beauty' figure

WHAT IT ENCODES

Dihydropyridine CCBs (amlodipine 5–10 mg, felodipine, long-acting nifedipine) are add-on antianginals that cause arterial/coronary vasodilation, reducing afterload and MVO₂. Long-acting DHPs are preferred (smooth, sustained effect). They are first choice for vasospastic (Prinzmetal) angina and useful when beta-blockers are contraindicated or angina persists despite them.

BOARD TRAP

WRONG: dihydropyridine CCBs slow the heart rate and shouldn't be combined with beta-blockers. Discriminator: DHPs (amlodipine) have minimal AV-node effect — they don't slow HR (may cause reflex tachycardia) and combine SAFELY with beta-blockers; it's the NON-DHP CCBs (verapamil/diltiazem) that are dangerous with beta-blockers.

7. Amlodipine safe with BB

WHAT YOU SEE

Beta-blocker soldier + amlodipine 'SAFE WITH BB'

WHAT IT ENCODES

Long-acting DHP CCBs (amlodipine) combine SAFELY with beta-blockers because they have negligible effect on the SA/AV node — the beta-blocker conveniently blunts any reflex tachycardia from the DHP's vasodilation, making the pairing complementary. This is a common, effective two-drug antianginal regimen when monotherapy fails.

BOARD TRAP

WRONG: adding amlodipine to a beta-blocker risks heart block and bradycardia. Discriminator: DHP + beta-blocker is a safe, synergistic combination (the beta-blocker offsets reflex tachycardia) — the heart-block danger belongs to verapamil/diltiazem + beta-blocker, not amlodipine.

8. Avoid short-acting nifedipine

WHAT YOU SEE

'AVOID short-acting nifedipine — reflex tachycardia / sympathetic activation — PRECIPITATES MI'

WHAT IT ENCODES

Immediate-release (short-acting) nifedipine causes rapid vasodilation and a surge of reflex sympathetic activation → tachycardia and ↑MVO₂ that can WORSEN ischemia and precipitate MI; it increased mortality in trials. Always use long-acting DHP formulations in IHD. (Short-acting nifedipine is also no longer used for hypertensive urgency for the same reason.)

BOARD TRAP

WRONG: short-acting nifedipine is a good rapid-acting antianginal/antihypertensive. Discriminator: immediate-release nifedipine causes reflex tachycardia and can precipitate MI (harm, not benefit) — only long-acting DHPs are used in IHD.

9. Non-DHP CCB + BB danger

WHAT YOU SEE

Verapamil knight + beta-blocker soldier with red X, 'AV BLOCK / BRADYCARDIA / HF'

WHAT IT ENCODES

Non-dihydropyridine CCBs (verapamil, diltiazem) act on the SA/AV node to slow HR and depress contractility. Combined with a beta-blocker their effects are ADDITIVE → high risk of severe bradycardia, high-grade AV block, and decompensated heart failure. Avoid this combination; non-DHPs are also contraindicated in HFrEF because of their negative inotropy.

BOARD TRAP

WRONG: verapamil or diltiazem can be safely added to a beta-blocker for better rate control in angina. Discriminator: non-DHP CCB + beta-blocker → additive AV block/bradycardia/HF — this combination is dangerous; if a patient is on a beta-blocker and needs a CCB, use a DHP (amlodipine).

10. Verapamil = rebound if stopped

WHAT YOU SEE

Verapamil 'NON-DHP', 'REBOUND ↑HR ↑MVO₂, ISCHEMIA/MI'

WHAT IT ENCODES

Non-DHP CCBs (verapamil/diltiazem) lower HR and contractility for rate control and are an alternative antianginal when beta-blockers are contraindicated (e.g., severe asthma) — but in HFrEF they worsen failure, and like beta-blockers their abrupt cessation can cause rebound ↑HR/↑MVO₂ and ischemia. Verapamil also causes constipation and can raise digoxin levels.

BOARD TRAP

WRONG: non-DHP CCBs are safe and beneficial in patients with reduced-EF heart failure. Discriminator: verapamil/diltiazem are negative inotropes contraindicated in HFrEF (they worsen decompensation) — only DHPs (amlodipine/felodipine) are tolerated in HFrEF if a CCB is needed.

11. Ranolazine (add-on)

WHAT YOU SEE

'RANOLAZINE FLASK', ranolazine pharmacist

WHAT IT ENCODES

Ranolazine inhibits the late inward sodium current, reducing Na/Ca overload and improving diastolic relaxation/coronary flow — an antianginal that does NOT lower heart rate or blood pressure. This makes it ideal as add-on therapy when angina persists but HR/BP are already low and limit beta-blockers, nitrates, or CCBs. It has no proven mortality benefit.

BOARD TRAP

WRONG: ranolazine works by lowering heart rate and blood pressure like a beta-blocker. Discriminator: ranolazine's distinguishing feature is NO effect on HR or BP (it blocks the late Na current) — that's exactly why it's chosen when low HR/BP preclude other antianginals.

12. Ranolazine QT / interactions

WHAT YOU SEE

'CYP3A chrome bumper', 'QT PROLONGATION', 'INTERACTION DANGER ZONE'

WHAT IT ENCODES

Ranolazine is metabolized by CYP3A4, so strong CYP3A inhibitors (diltiazem, verapamil, ketoconazole, macrolides, grapefruit) raise its levels — and it prolongs the QT interval. Avoid with other QT-prolonging drugs and in significant hepatic impairment; reduce the dose with moderate CYP3A inhibitors. Despite QT prolongation it has not been shown to cause torsades clinically.

BOARD TRAP

WRONG: ranolazine is free of significant drug interactions and QT effects. Discriminator: ranolazine is a CYP3A substrate that prolongs QT — watch for interactions (verapamil/diltiazem, azoles, macrolides, grapefruit) and avoid stacking QT-prolonging drugs.

13. Ivabradine (if EF normal, sinus)

WHAT YOU SEE

'IVABRADINE when EF is normal', 'SINUS NODE', 'OK HR >70 + EF'

WHAT IT ENCODES

Ivabradine selectively inhibits the SA-node 'funny' (If) current, lowering heart rate WITHOUT affecting contractility or blood pressure. It is an add-on antianginal for patients in SINUS rhythm with HR still >70 despite a maximally tolerated beta-blocker, or when beta-blockers are not tolerated. Side effects: visual phosphenes (luminous brightness) and bradycardia. It has a role in chronic HFrEF (EF $\leq$ 35%, sinus, HR $\geq$ 70).

BOARD TRAP

WRONG: ivabradine works in atrial fibrillation and also boosts contractility/blood pressure. Discriminator: ivabradine only works in SINUS rhythm (it acts on the SA node — useless in AF) and purely slows HR with no inotropic or BP effect.
